

Sachin Chandran

Ph.D. Candidate, Department of Mathematics, Oklahoma State University
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RESEARCH INTERESTS

My research lies in Lie theory and the representation theory of reductive groups. My doctoral work studies the geometric relationship between the representations of p -adic groups and real reductive groups, in the spirit of the Adams–Barbasch–Vogan framework and the local Langlands correspondence.

EDUCATION

Ph.D. in Mathematics **2021–present**

Oklahoma State University, Stillwater, OK, USA

Advisor: Prof. Leticia Barchini

Dissertation: *A geometric comparison of Iwahori spherical representation of forms inner to $PGL(n, \mathbb{Q}_p)$ and irreducible representation of real forms of $PGL(n, \mathbb{C})$*

Expected completion: July, 2027

Integrated BS–MS (Mathematics major) **2014–2019**

Indian Institute of Science Education and Research (IISER) Mohali, India

Master’s thesis: *The study of classical groups as linear algebraic groups.*

Advisor: Dr. Tanusree Khandai

CONFERENCES AND WORKSHOPS ATTENDED

Arizona Winter School: Representation of p -adic groups **2025**

36th Automorphic Forms Workshop **2024**

Texas–Oklahoma Representation Theory and Automorphic Forms **2023**

Southeastern Lie Theory Workshop **2022**

HONORS, AWARDS, AND FELLOWSHIPS

Schiller J. Scroggs Distinguished Graduate Fellowship, Oklahoma State University **2024–2025**

O. H. Hamilton Award, Oklahoma State University **2022–2023**

CSIR JRF–NET Fellowship, Government of India **2019–2020**

INSPIRE Fellowship, Department of Science & Technology, Government of India **2014–2019**

TEACHING EXPERIENCE

Instructor of Record **2022–present**

Oklahoma State University, Department of Mathematics

Instructor responsible for lectures, assessment, and grading for the following sections:

- Calculus I — Fall 2025; Spring 2025, Fall 2024, Spring 2024, Fall 2023; Spring 2023
- Calculus I (online, asynchronous) — Summer 2026, Spring 2026, Summer 2025
- Differential Equations — Summer 2023

- Preparation for Calculus — Spring 2022, Fall 2022

Guest Lecturer

2019–2020

University of Calicut, School of Distance Education, India

Prepared lecture notes and delivered undergraduate courses including Linear Algebra, Calculus, Probability Theory, Complex Analysis, Real Analysis, Introductory Number Theory, and Discrete Mathematics.

SERVICE AND LEADERSHIP

Liaison, Mathematics Graduate Student Society, Oklahoma State University

2024–2025

Sports Coordinator, International Student Organization, Oklahoma State University

2024–2025

President, Badminton Club, Oklahoma State University

2023–2024

TECHNICAL SKILLS

Programming C, C++, Python, SageMath, Octave

Typesetting $\text{\LaTeX}$